

8T SRAM Cell as a Multibit Dot-Product Engine for Beyond Von Neumann Computing



IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS,
VOL. 27, NO. 11, NOVEMBER 2019

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Jul. 3rd 2025



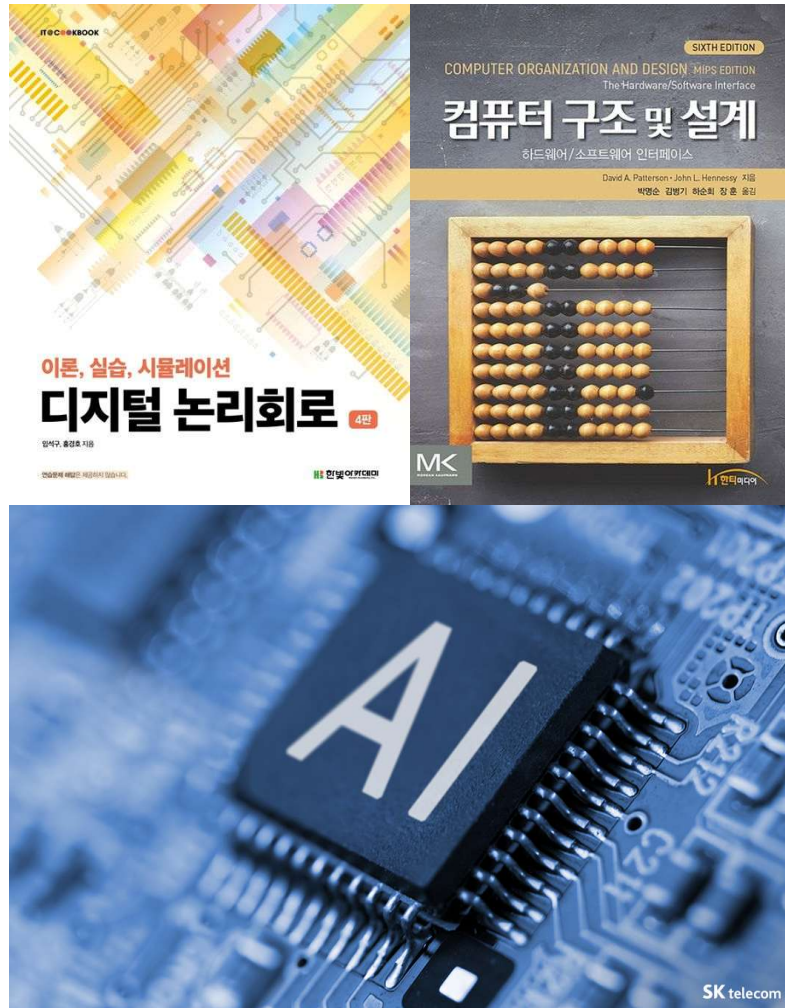
8개의 트랜지스터로 만든 SRAM(캐시)로

Multibit Dot-Product 연산을 수행하여

8T SRAM Cell as a Multibit Dot-Product Engine for Beyond Von Neumann Computing

폰노이만 구조를 극복하는 논문

Overview



■ 논리회로 & 컴퓨터구조

- MOSFET Transistor 동작
- Transistor 기반 Device 설계

■ Next gen Computer Device

- MultiDotEngine in SRAM
- How to Overcome bottleneck?



컴퓨터가 어떻게 만들어지는지
그리고 어떻게 변화할지를 이해!



Electronic Switch

Transistor

Transistor



- Resistor

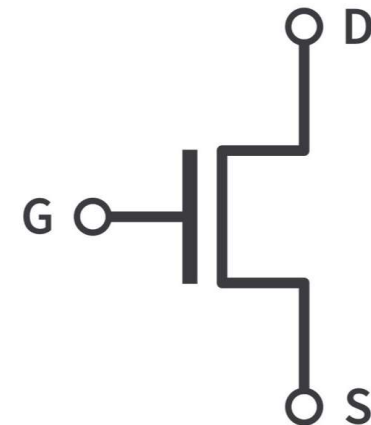
- 2단자 소자 (IN/OUT)



Resistor

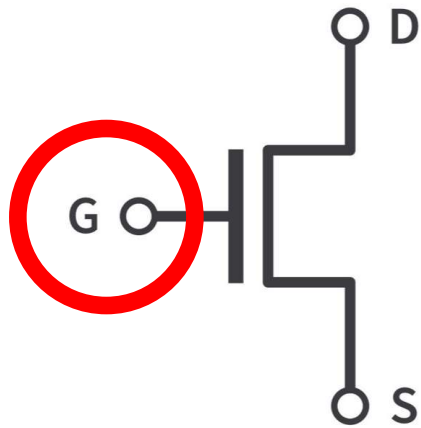
- Transistor

- 3단자 소자
- **Gate**, Drain, Source로 구성



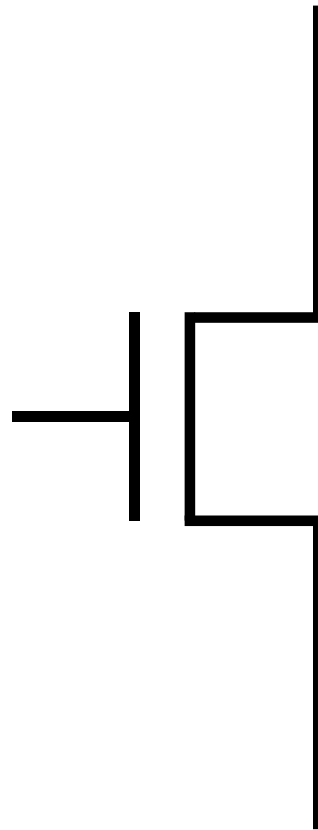
Transistor

Transistor

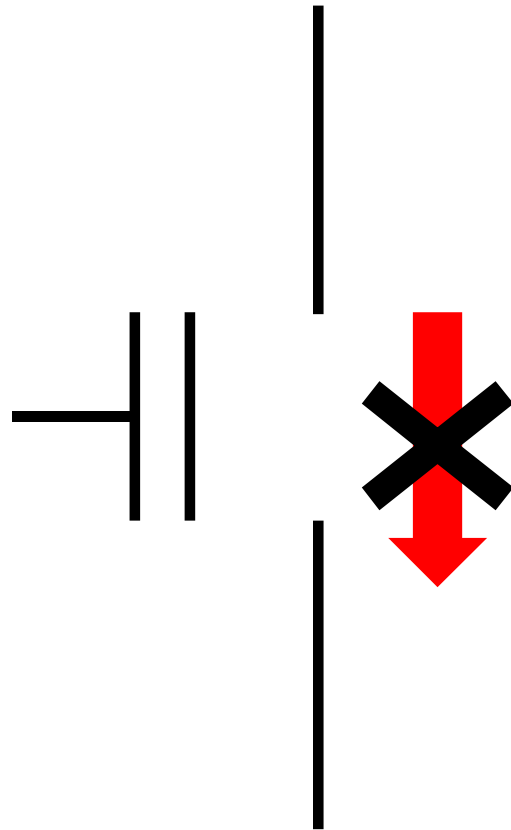


- Gate(G)에 전압을 인가하면 전류가 흐르는 통로가 생김
- 마치 스위치와 같은 역할을 함

Transistor



Transistor

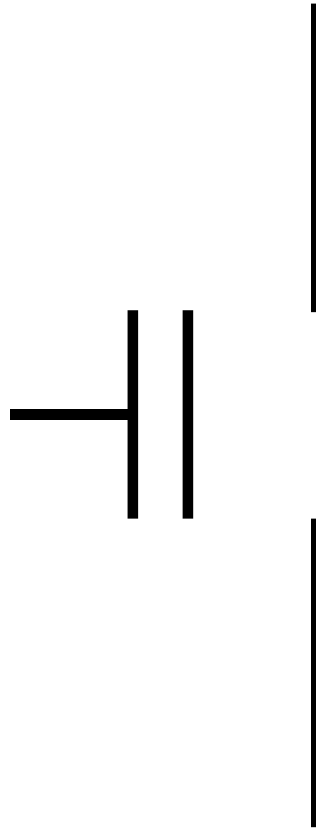


기본 상태에서는
Transistor에 전류가
흐르지 못함

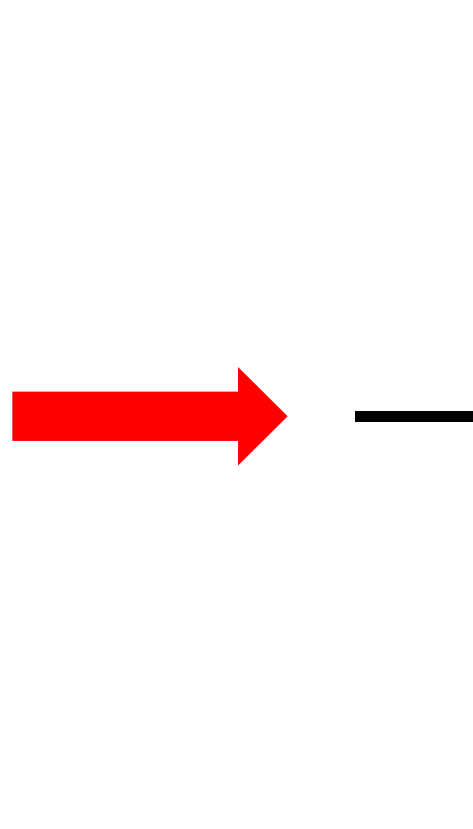
Transistor



Gate에 전압 인가!



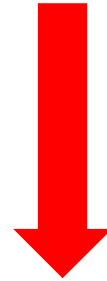
Transistor



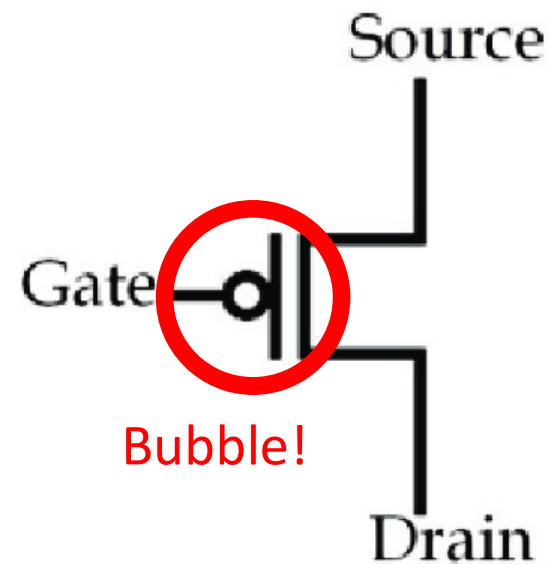
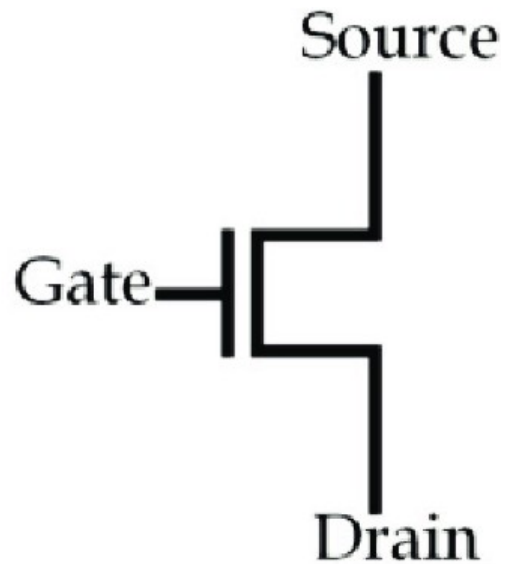
Transistor



이제 전류가
정상적으로 흐른다!



Transistor Type: MOSFET



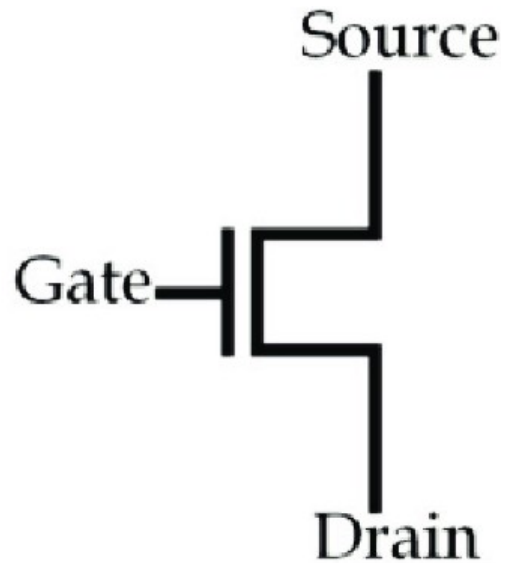
Transistor Type: MOSFET



* 참고

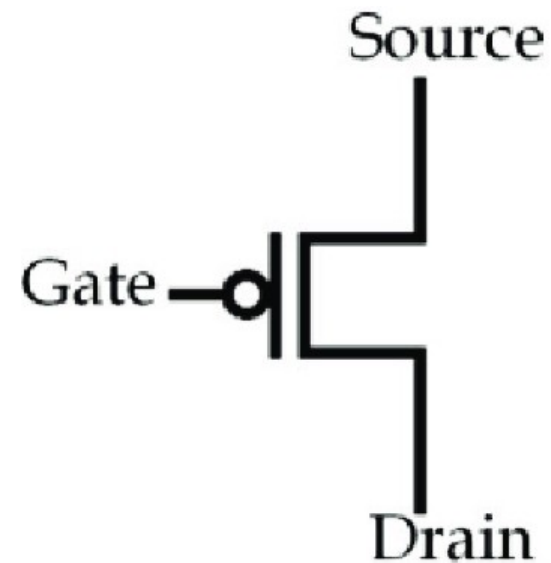
HIGH: 5V

LOW: 0V



NMOS

Gate에 "HIGH" 전압 인가



PMOS

Gate에 "LOW" 전압 인가

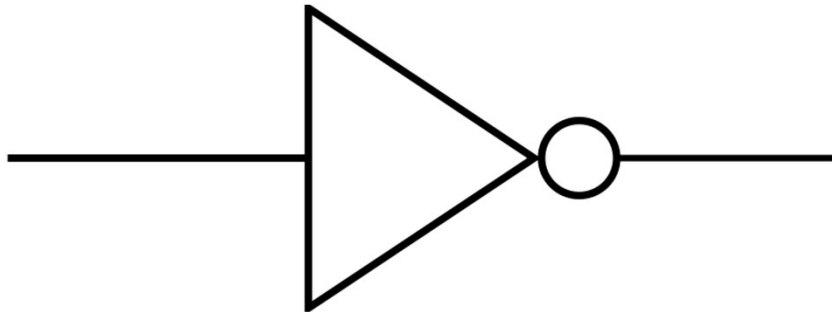


With MOSFET

Logic Circuit

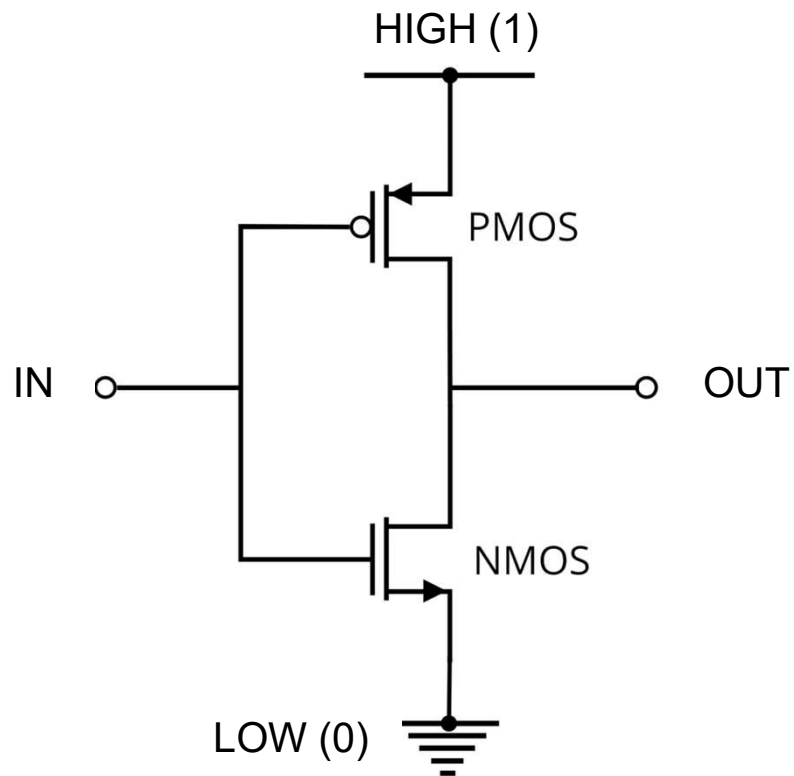
When you were 1st year student..

Logic Circuit: Inverter



- Inverter
 - 논리회로 수업에서 배우는 논리소자
 - 0을 1로, 1을 0으로 반전
- MOSFET Transistor 2개로 제작

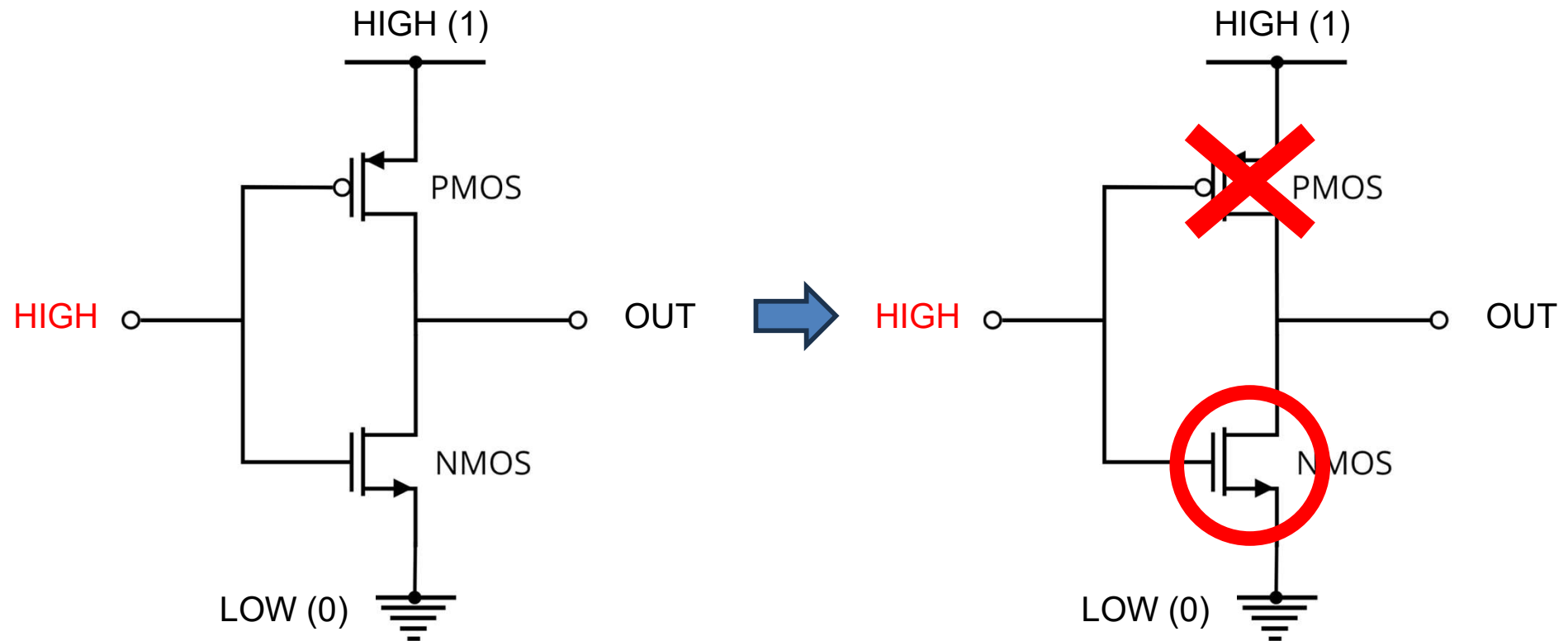
Logic Circuit: Inverter



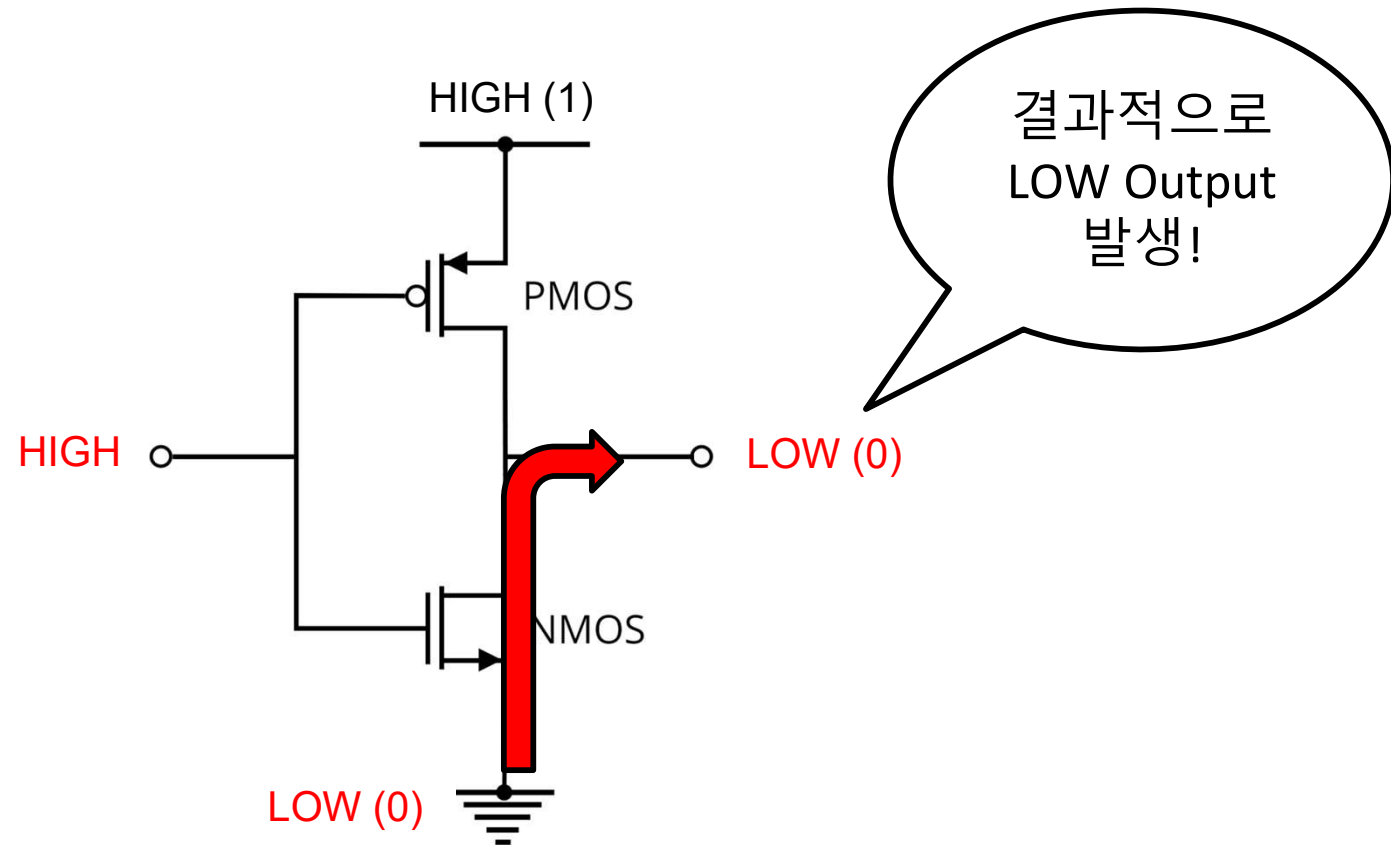
■ Inverter Schematic

- NMOS와 PMOS의 직렬연결
- 입력과 출력을 공유

Logic Circuit: Inverter



Logic Circuit: Inverter

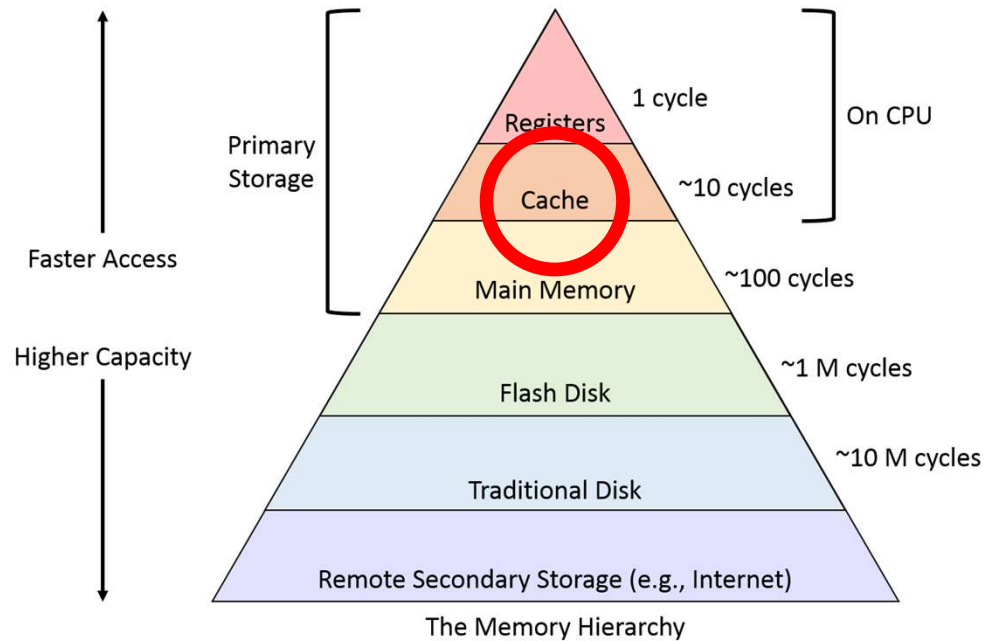




With MOSFET

Computer Device

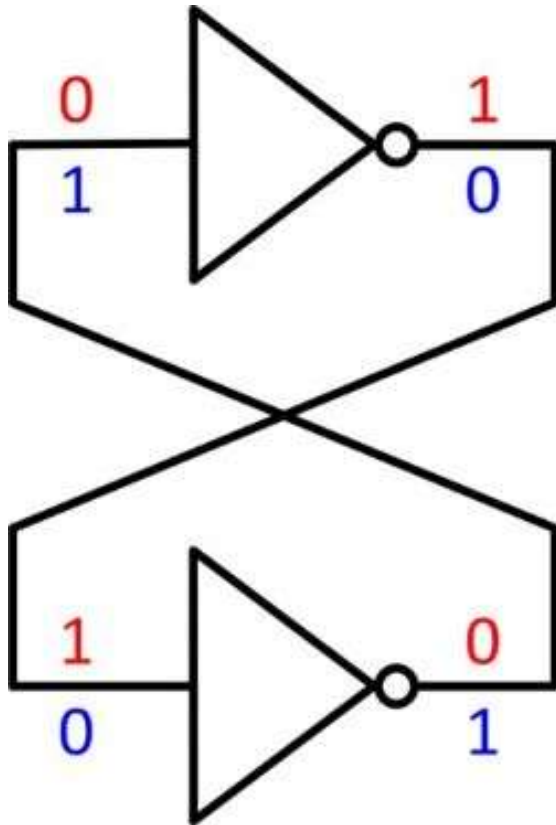
Computer Device: SRAM



■ SRAM

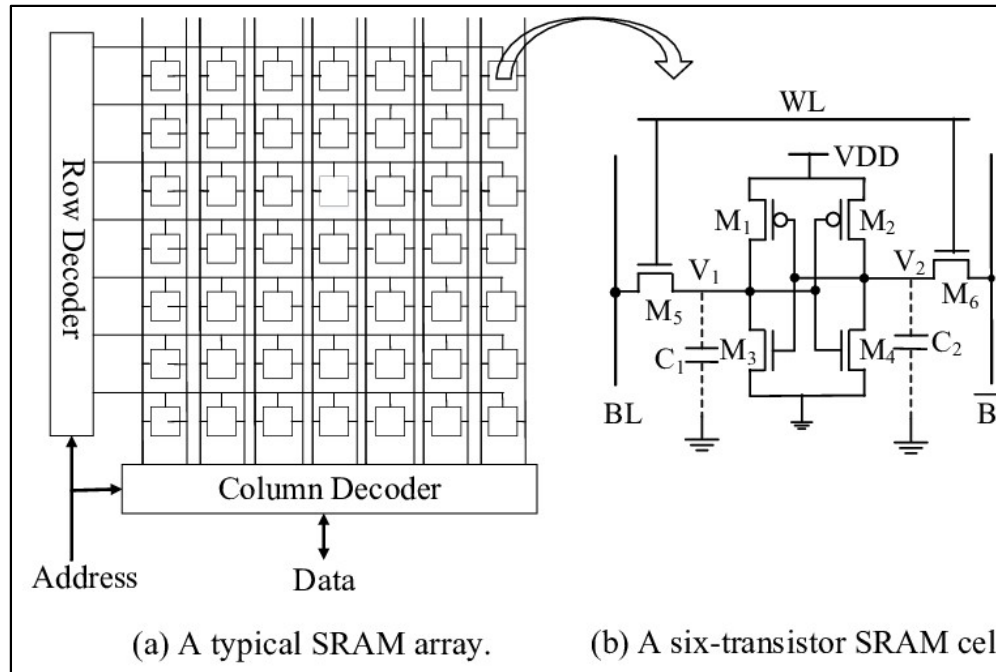
- 컴퓨터에서 Cache로 사용
- 최소 6개의 TR로 제작

Computer Device: SRAM

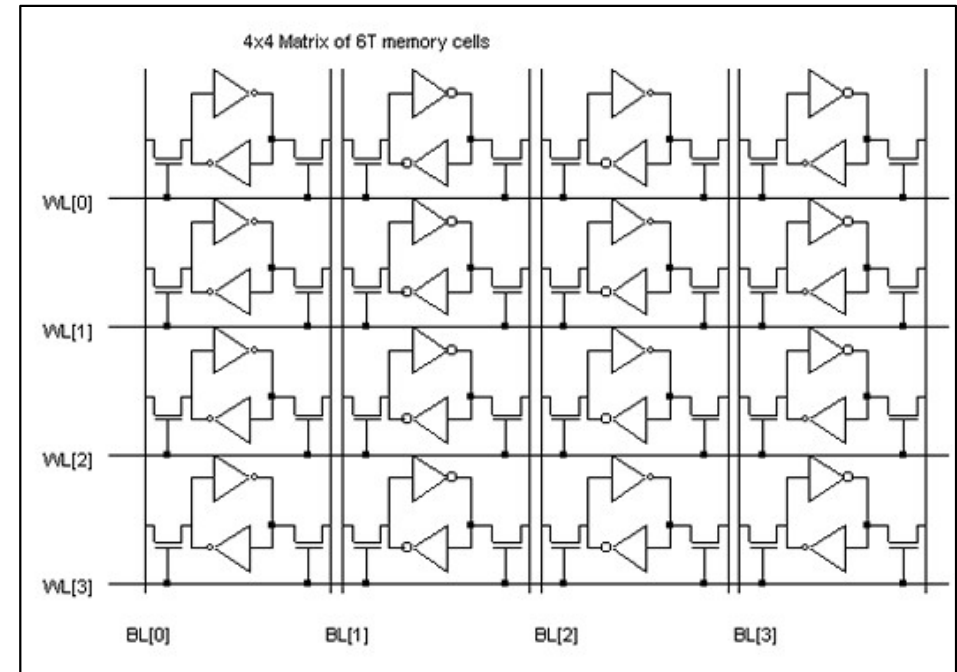


- Cross Coupled Inverter
 - 두 인버터가 서로 연결된 구조
(한 쪽 출력이 다른 쪽의 입력)
 - 전하를 효과적으로 저장

SRAM Cell Array



SRAM Array Schematic



Symbol Schematic

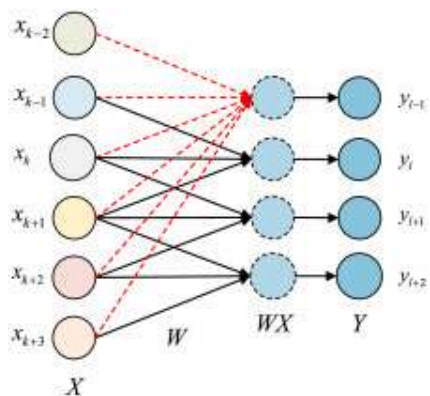
Cell 구조의 array로 SRAM 완성!



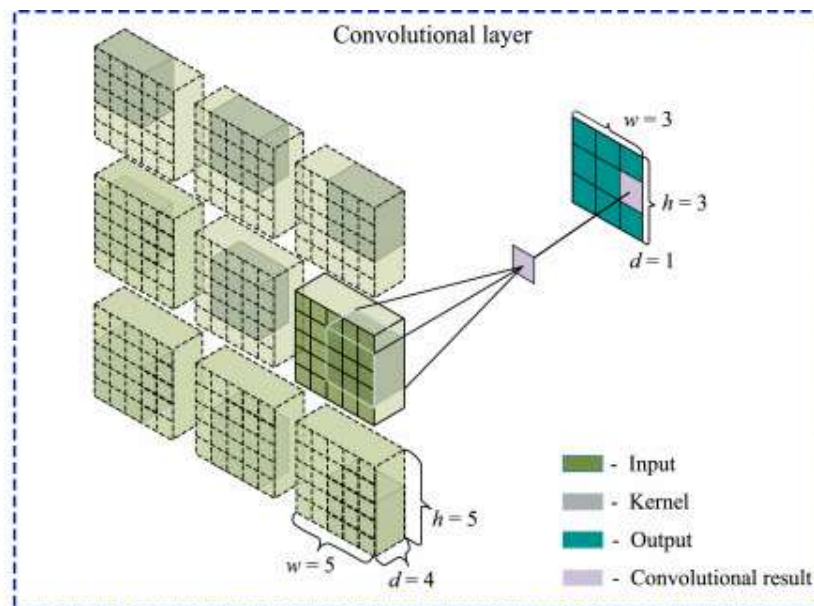
Next Gen SRAM

Multi-Dot Product Engine

Age of AI

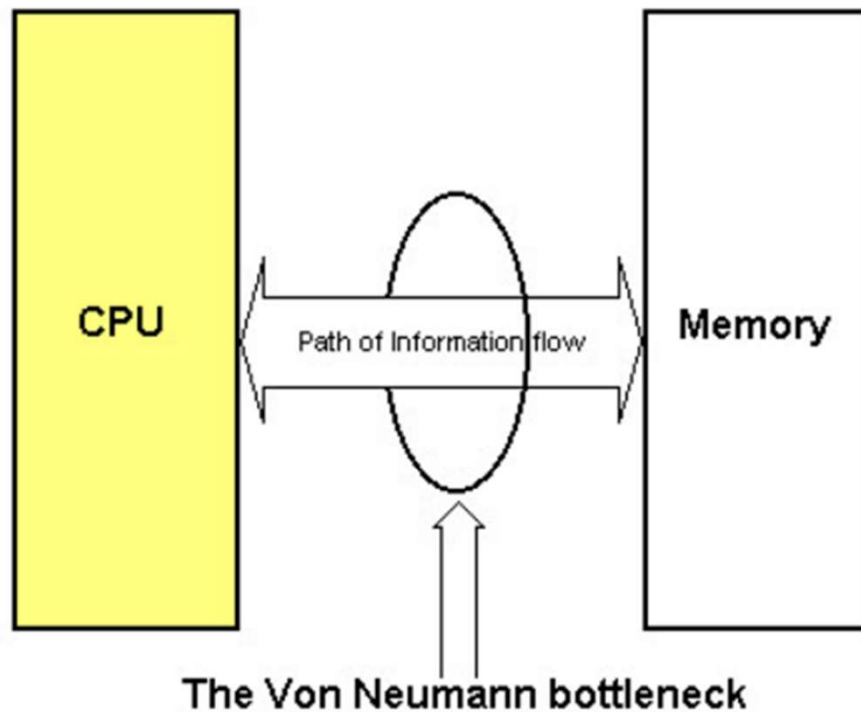


(a)



(b)

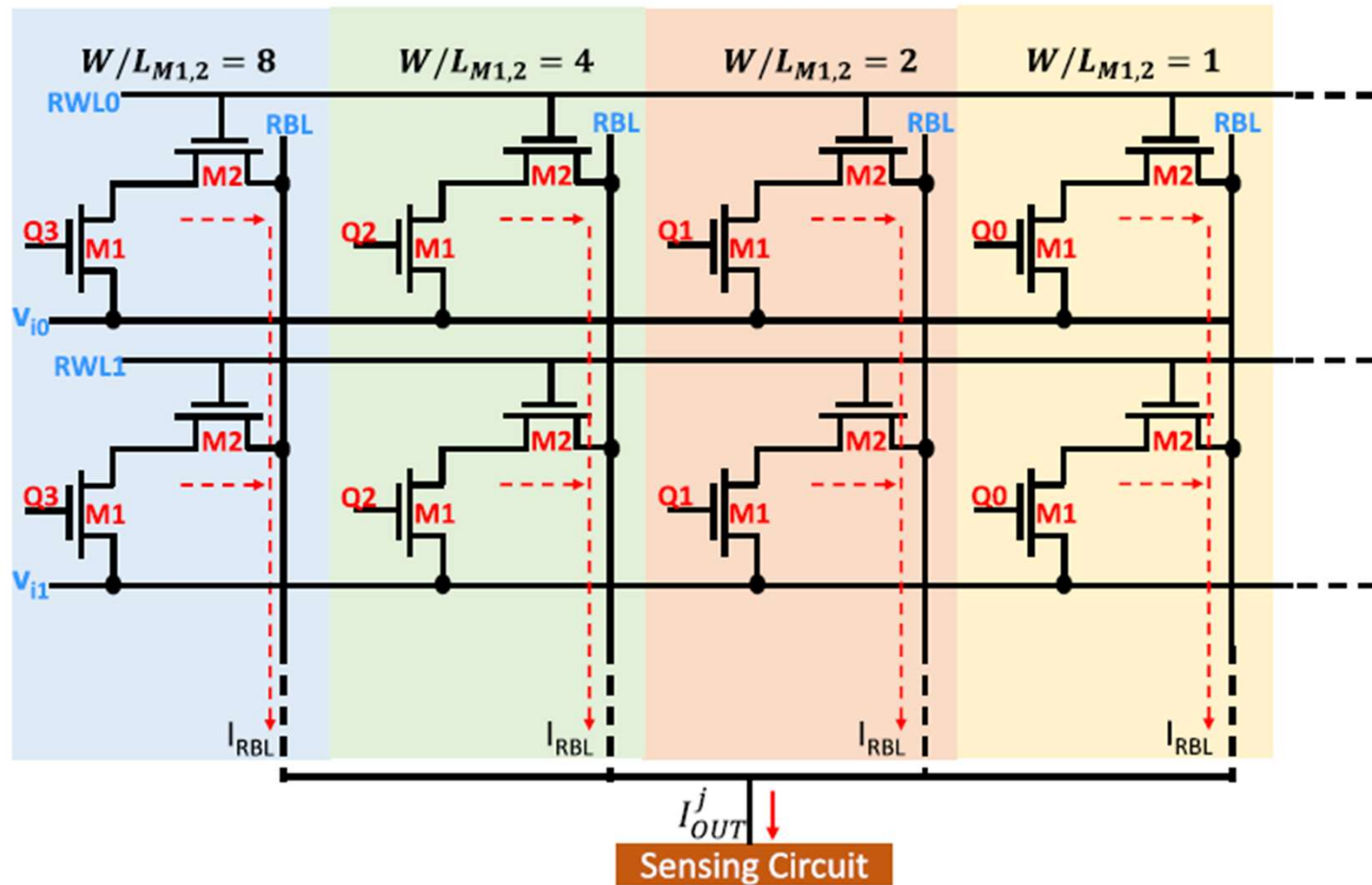
Machine Learning & Deep Learning은
많은 양의 **Dot Product 연산**이 필요!



- Multi-Dot Product
 - 연산 자체의 난이도는 단순
 - But 연산의 양이 많은 편

-> Von Neumann Bottleneck 발생!
- 논문의 목표
 - 메모리 내에서 연산을 할 수 있을까?

Multidot Product Engine in SRAM



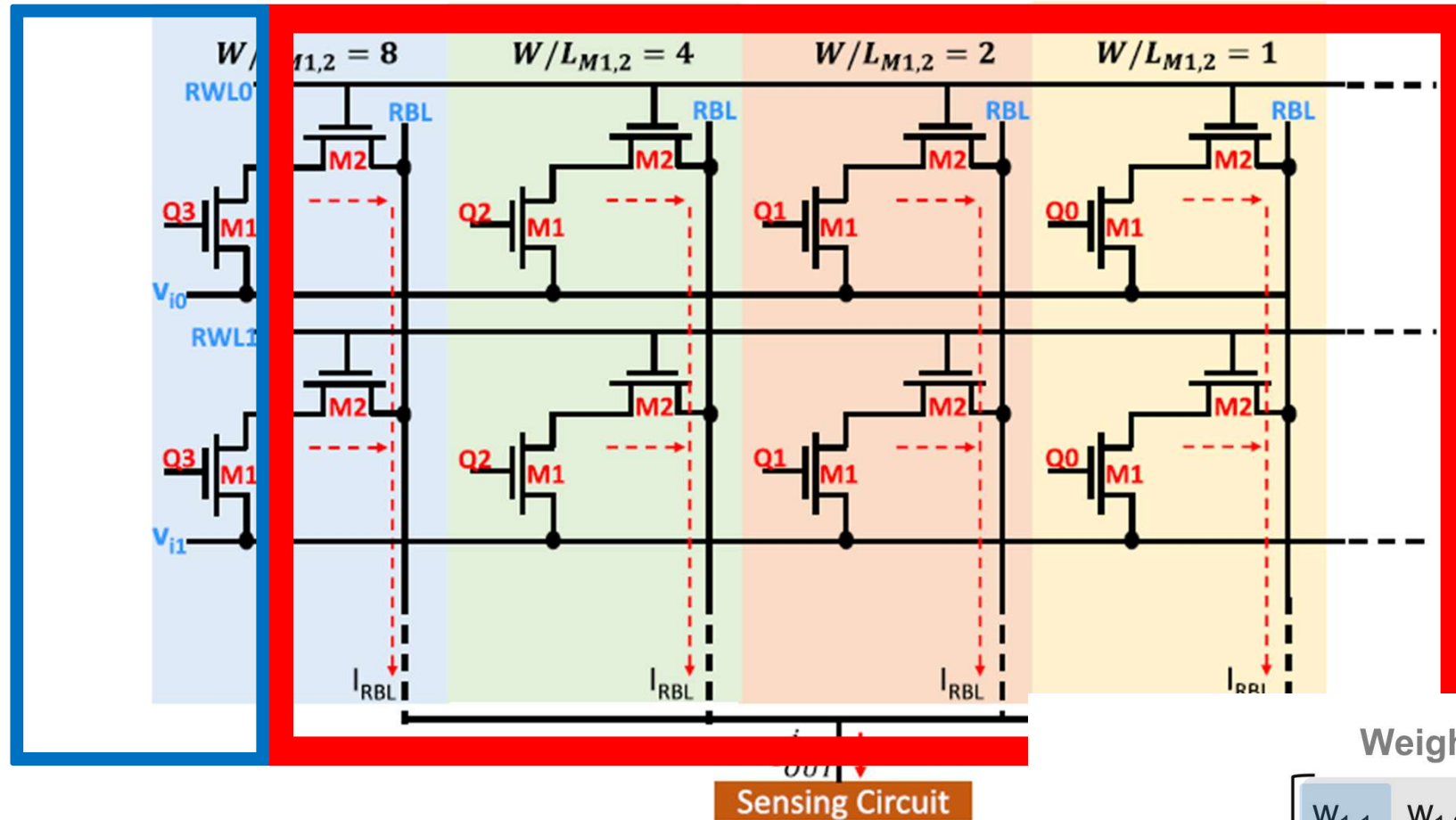
Multidot Product SRAM Structure

Multidot Product Engine in SRAM



Input Vector

Weight Matrix



Sensing Circuit

Weight matrix

$w_{1,1}$	$w_{1,2}$	$\dots$	$w_{1,m}$
$w_{2,1}$	$w_{2,2}$	$\dots$	$w_{2,m}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$w_{n,1}$	$w_{n,2}$	$\dots$	$w_{n,m}$

$[p_1 \ p_2 \ \dots \ p_n]$

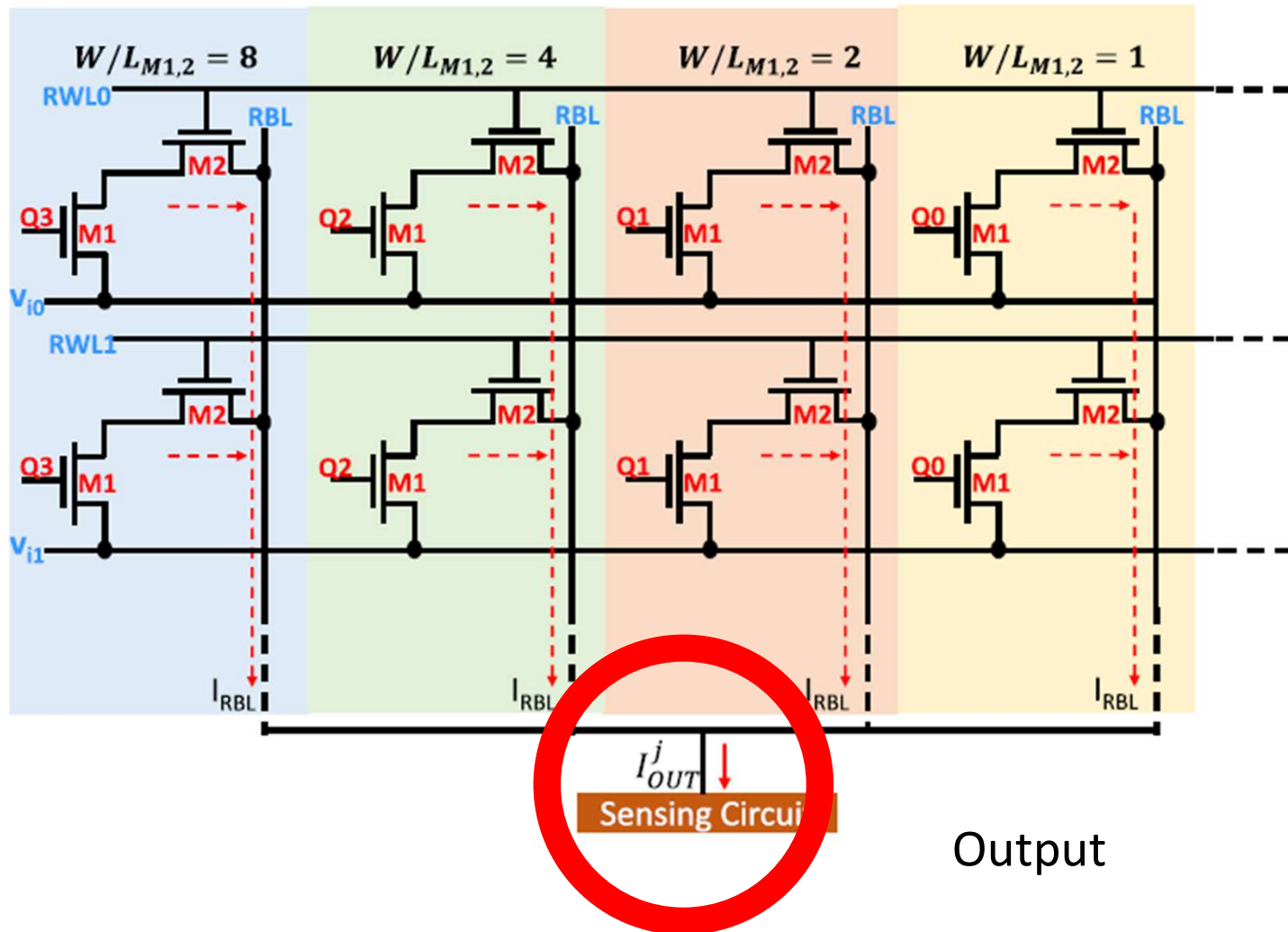
Input vector

Multidot Product Engine in SRAM



Input Vector

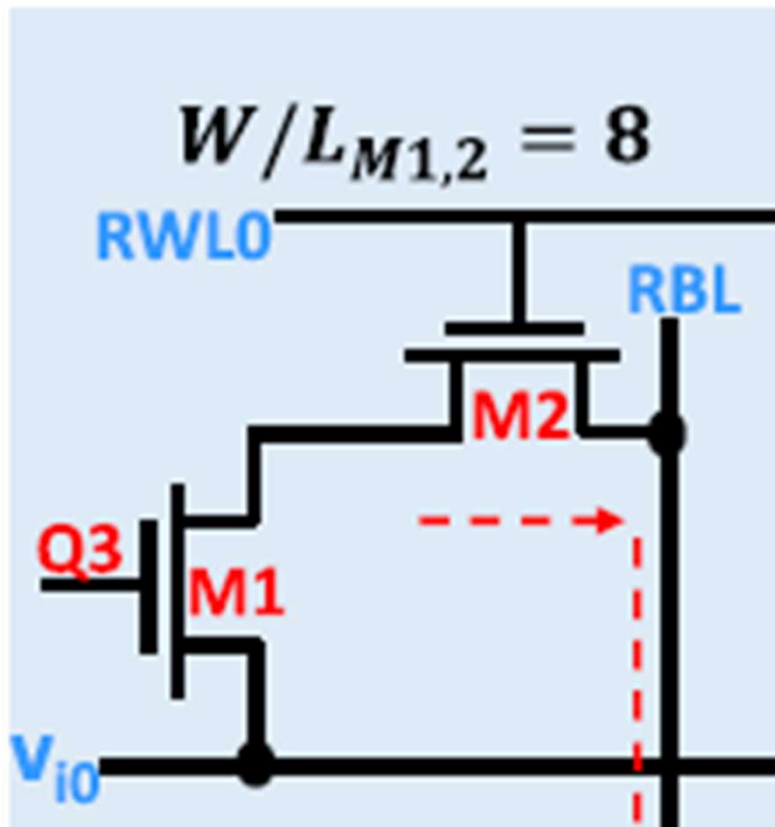
Weight Matrix



Output

각 열의 전류들의 합

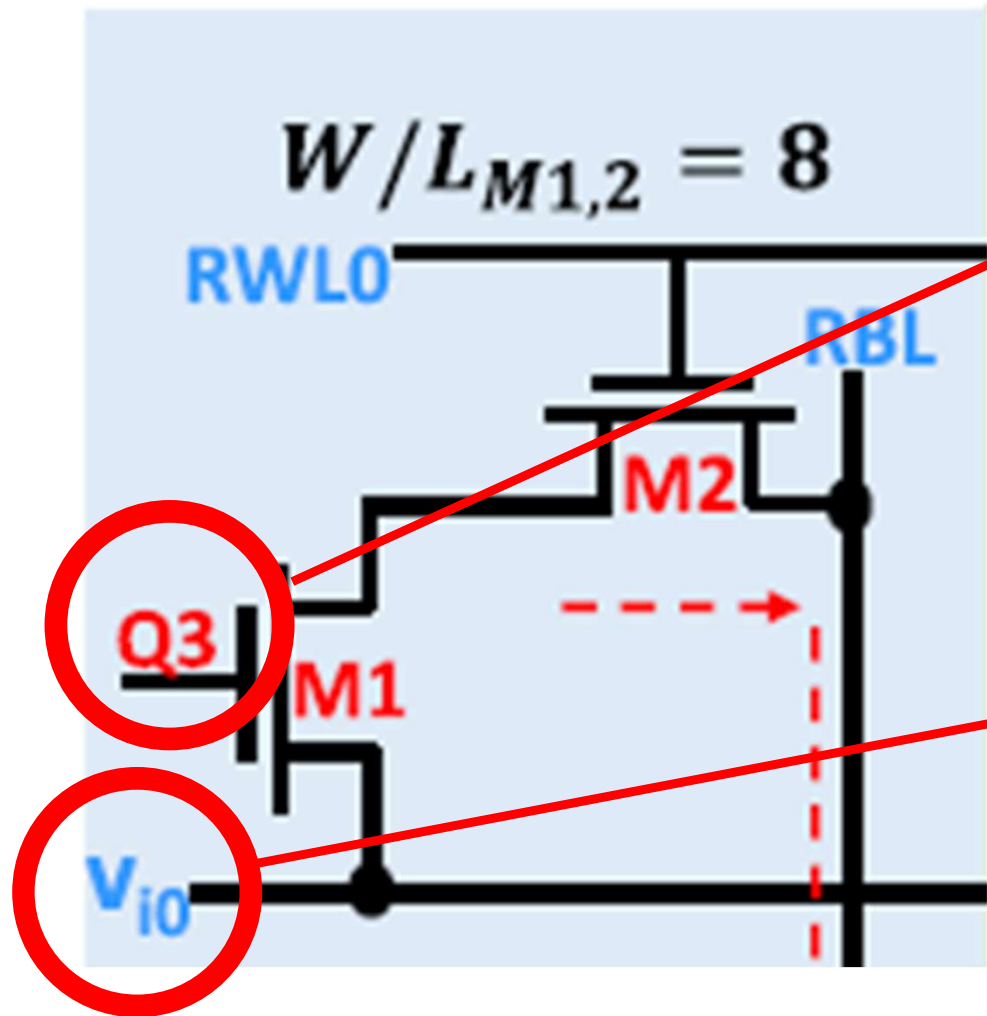
Multidot Product Engine in SRAM



$$y = \sum_{i=0} w_i * x_i$$

$$\begin{aligned} \Sigma(v_i \cdot W_i^j) &= \Sigma[v_i \cdot (2^3 w_3 + 2^2 w_2 + 2^1 w_1 + w_0)] \\ &= \Sigma(v_i \cdot 2^3 w_3) + \Sigma(v_i \cdot 2^2 w_2) \\ &\quad + \Sigma(v_i \cdot 2^1 w_1) + \Sigma(v_i \cdot w_0). \end{aligned}$$

Multidot Product Engine in SRAM



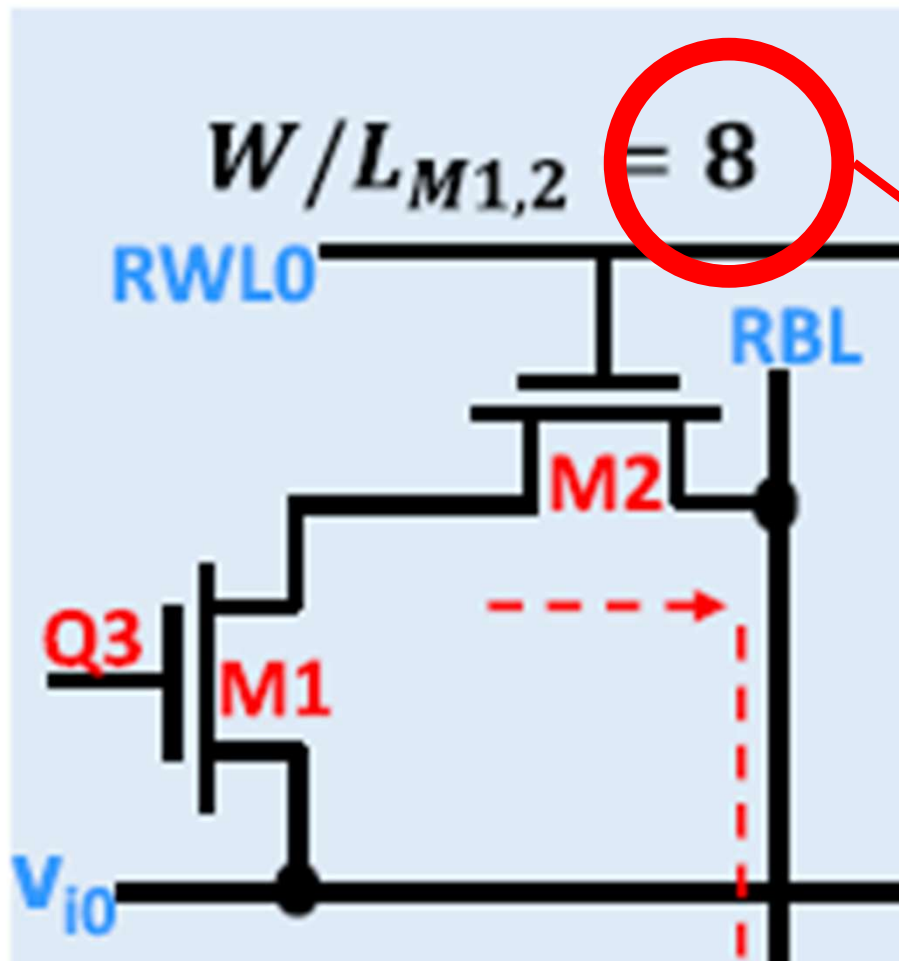
1. Binary Weight

각 SRAM Cell에 저장된
0 또는 1의 값

2. Input Voltage

아날로그 입력전압
(= Input Vector)

Multidot Product Engine in SRAM



3. TR Width

각 열의 TR의 Width를 조절하여
Matrix의 Column을 구분

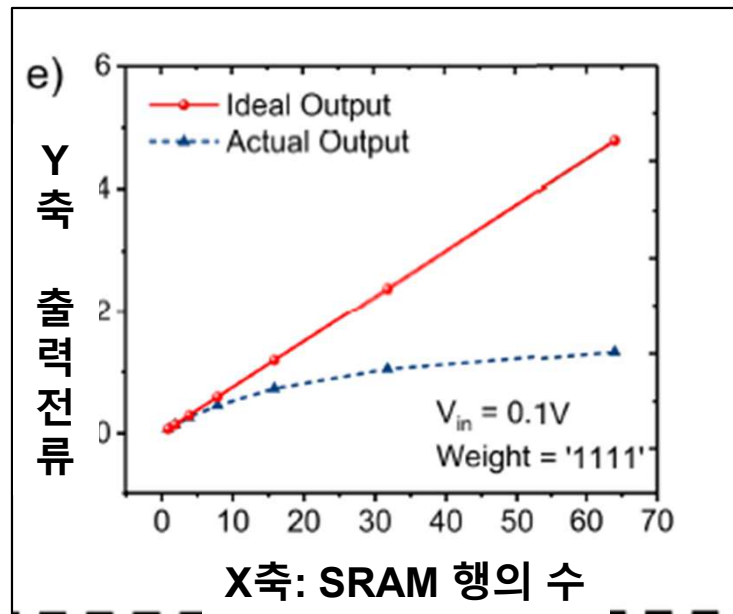


For Multidot SRAM

Peripheral Amplifier

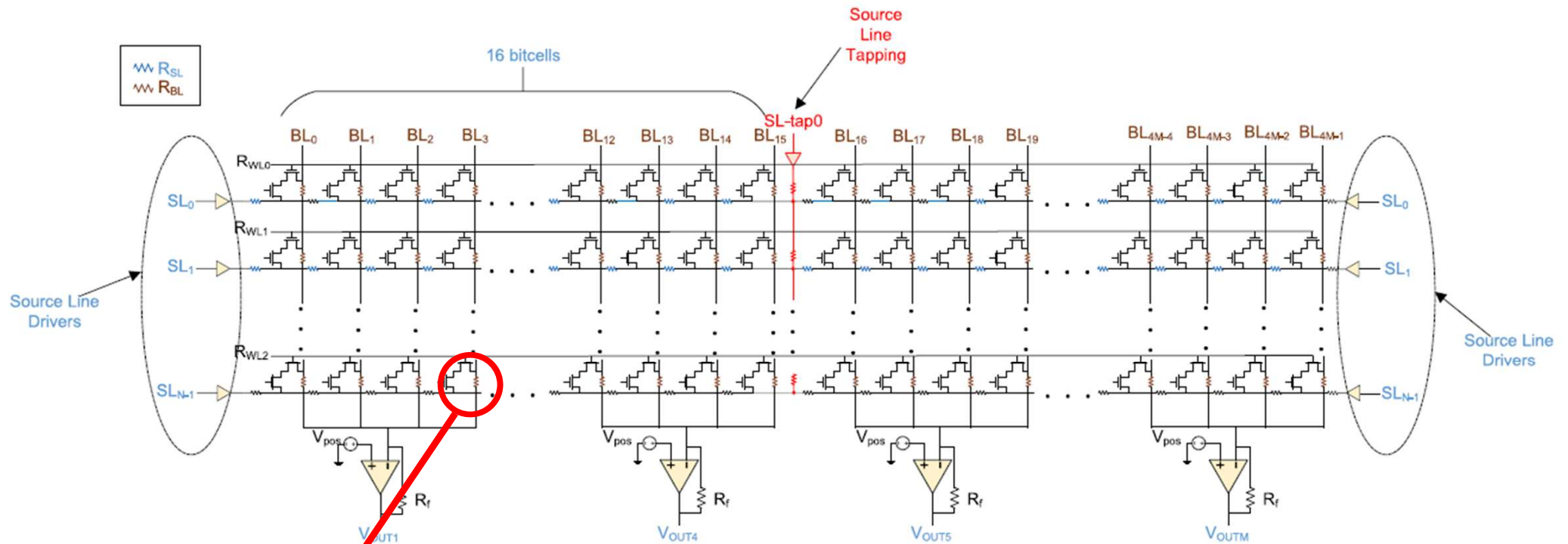
Reality is different

Peripheral Amplifier



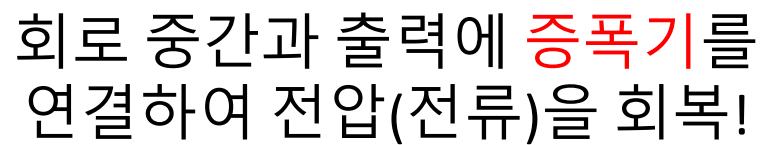
- Ideal VS Real
 - Simulation 결과 출력 전류가 떨어지는 것을 확인
 - SRAM Array가 커질수록 심해짐

Peripheral Amplifier



기생저항

- Nonideal Output의 원인
 - Array가 커질수록 전선이 길어짐
 - 전류가 전선을 따라 흐르며 감소





So... is it useful?

Advantage of Multidot SRAM

Advantage of Multidot SRAM



TABLE I
COMPARISON OF 8T DPE INFERENCE ACCURACY ON MNIST

Network Configuration	Accuracy (%)
Ideal (Software)	98.27
8T DPE (No VT variations)	98.16
8T DPE (With VT variations)	98.15

1. High Accuracy

MNIST Dataset Dot-Product Accuracy가 거의 Ideal

2. Powerful Vector Calculation

ML/DL이 아닌 일반적인 Vector 연산도 빨라짐

Advantage of Multidot SRAM

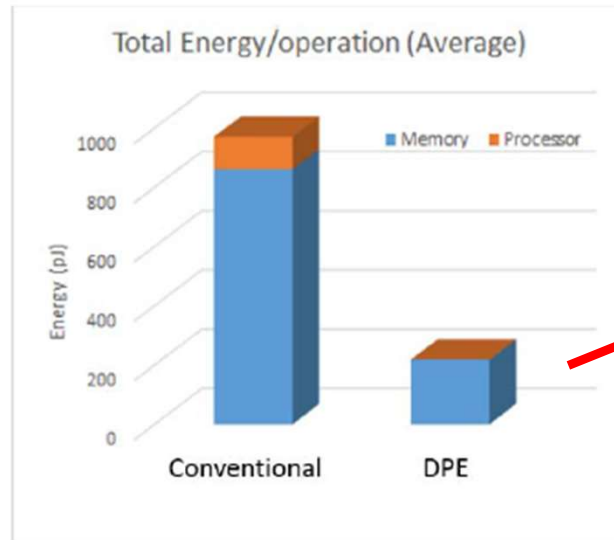


Fig. 13. Average energy comparison between conventional digital sequential implementation and the proposed DPE. The energy is reported for 16×16 dot-product computations, wherein 16 rows are simultaneously activated and each row consists of 16 4-bit words.

기존의 Computing 대비
80%의 전력 감소!

3. High Speed and Energy Efficiency
속도뿐만 아니라 전력 효율도 상승

Conclusion



- Computer Device의 설계
 - Transistor를 통해 다양한 논리 회로 설계
 - 이들을 합쳐 Computer Device 설계
- 전통적인 컴퓨터 구조의 변화
 - 기존의 폰 노이만 구조에서 인메모리 구조로의 변화
 - Non ideal한 회로의 한계를 극복하는 과정
 - 빅데이터 처리를 위해 SW뿐만 아니라 HW의 역할 또한 중요

Thank you!

Questions?